FEDERAL SPECIFICATION

DEVELOPER, PHOTOGRAPHIC, LOW ACTIVITY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for a low activity, medium grain developer composed of mixed chemicals packaged in powder form for use in processing black and white photographic film to medium contrasts.

1.2 Classification.

1.2.1 *Sizes.* The developer shall be furnished in two sizes, containing sufficient powder to prepare one gallon and three and one-half gallons of working solution, respectively, (see 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

O-C-275—Chemicals, Photographic "Single Substance", General Specification for.

PPP-B-636—Box, Fiberboard.

PPP-C-96—Cans, Metal, 28 Gage and Lighter.

PPP-P-16—Packaging of Photographic Dry Chemicals, General Specification for.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

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American Standards Association, Inc., (ASA) Standards:

PH4.125—Mono-Methyl-Para-Aminophenol Sulfate.

PH4.126—Hydroquinone.

PH4.230—Sodium Tetraborate, Decahydrate (Borax).

PH4.275—Sodium Sulfite, Anhydrous.

PH4.1—Manual Processing of Black-and-White Photographic Films and Plates, Methods for.

(Copies of ASA standards may be obtained from the American Standards Association, Inc., 10 East 40th Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Materials. Photographically active components used shall be as specified herein. Materials used to prevent deterioration of the dry chemicals which are not specifically designated shall be entirely suitable for the purpose intended. The materials used shall be freely soluble in water and relatively nontoxic.

3.2 Data. No data is required by this specification or by applicable publications referenced in section 2, unless specified in the contract or order (see 6.2).

3.3 Chemicals. The chemicals used shall be photographic grade conforming to O-C-275, as applicable, or the appropriate ASA standard. If neither a government specification nor ASA standard is applicable, then a grade of chemical sufficiently pure for photographic use shall be selected.

3.3.1 *Impurities.* Impurities and dirt in the combined ingredients of the packaged developer powder shall not exceed the cumulative tolerances for the ingredients as specified in the appropriate ASA standards.

3.3.2 *Formula.* The developer powder shall contain the chemicals specified in table I in the proportions shown to make one gallon or three and one-half gallon(s) of working solution. This specification calls for either a single-powder preparation or a two-powder preparation.

TABLE I.

Chemical name	ASA standard	To make one liter	Tolerance (plus or minus)
		Grams	*Percent*
Mono-methyl-para-aminophenol Sulfate	PH4.125	2.0	8
Sodium Sulfite, Dessicated	PH4.275	100.0	5
Hydroquinone	PH4.126	5.0	8
Sodium Hexametaphosphate	None	0.5	20
Sodium Tetraborate, Decahydrate (Borax)	PH4.230	2.0	8

3.3.3 *Tolerances.* The tolerances of the individual ingredients in each unit package shall not exceed those specified in table I. The tolerance of the total net weight of all the mixed chemicals in each container shall not exceed plus or minus three percent of the marked net weight.

3.3.4 *Additional ingredients.* Ingredients, in addition to those shown in table I, may be used to stabilize the chemicals, decrease the deterioration of the packaged powder and prolong the keeping qualities and useful life of the working solution. If additional ingredients are used, they shall be entirely suitable for the intended purpose and have no detrimental effects on the photographic performance of the developer or on the film processed in the developer solution. If additives are used, their weight shall be included in the total net weight marked on each unit container.

3.4 Physical properties. The dry chemicals used in the developer powder shall be suitably pulverized or granulated to eliminate lumps and ensure a free flowing mixture in the powder form. The chemicals shall be thoroughly blended so the powder, in each container, is a homogeneous mixture.

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3.4.1 *Color of powder.* The dry developer powder shall be light in color and free of brown coloration, dark specks, or other evidence of deterioration.

3.4.2 *Solubility and appearance.* The developer powder shall be readily soluble when dissolved according to the directions marked on the package. There shall be no dirt or insoluble precipitates when the solution is cooled to 68°F. The color and clarity shall be comparable to that of a freshly prepared solution mixed according to the formula specified in table I. There shall be no precipitation when the developer solution is cooled to 60°F.

3.4.3 *Alkalinity.* When tested in accordance with section 4, the pH of the solution prepared from the powder shall be within plus or minus 0.1 pH unit of a freshly prepared solution mixed according to the formula specified in table I.

3.5 **Sensitometric characteristics.** The sensitometric performance of the developer powder, when prepared according to the instructions for use on the label, shall match the performance of a freshly prepared developer, mixed according to the formula specified in table I, to pass the photographic test in section 4.

3.6 **Unit package.** The chemicals may be packaged as a single powder or as a two powder unit package when the developing agents, mono-methyl-para-aminophenol sulfate and hydroquinone, are placed within a separate container.

3.6.1 *Single powder package.* The unit package shall conform to the requirements of PPP-P-16, type II, class 1 and PPP-C-96, type 1, round metal cans. The can may be provided with a rip or scored tab for opening the can with the fingers.

3.6.1.1 *Can opener.* If a can opener is needed, one commercial, metal-puncture-type opener shall be included in each intermediate package.

3.6.2 *Two powder package.* The unit package shall consist of two containers referred to as inner and outer containers.

3.6.2.1 *Outer container.* The outer container shall conform to PPP-P-16, type II, class 1, round metal can and shall be provided with a key-wind, rip-type opening device and key.

3.6.2.2 *Inner container.* The inner container may be constructed of any suitable material. The material used shall be inert with respect to the chemicals used in the developer, or shall be coated both inside and outside with a suitable material which is inert to the chemicals and the container. The inner container shall be so constructed and closed that the chemicals therein will not come in contact with the chemicals in the outer container. The inner container shall be suitably marked to indicate that its contents will be dissolved first.

3.6.3 *Dimensions.* Dimensions of the containers shall be such that not less than 80 percent of the internal volume will be utilized and such that the contents will not spill when the containers are opened in an upright position. If regularly available commercial sizes meeting this requirement are not available, the requirements may be waived by the requiring agency if requested, with affidavits, by the supplier.

3.6.4 *Closure.* The completed unit package shall be waterproof and air-tight.

3.6.5 *Labeling.* Each unit package shall be durably and legibly labeled by lithographic or other suitable means. If gummed labels are used they shall be the wrap-around type. The labels shall be affixed in place with water resistant adhesive. The adhesive shall not smear or blur the markings. The following information shall be on the label.

a. Item identification.
b. Directions for mixing and use.
c. Manufacturer's name.
d. Final volume when mixed in U.S. gallon(s) and metric liters.
e. Net weight.

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3.7 **Storage life.** The developer powder, in the unit package, shall pass the storage life test and photographic test given in section 4, after storage for four weeks at 120° ± 3°F. The time required, for the developer solution prepared from the stored powder, to develop exposed film to a gamma of 0.9 ± 0.1 shall not exceed by more than one minute the time required for a freshly prepared developer solution to develop the same film to the same gamma.

3.8 **Affidavit.** The contractor shall furnish to the Government inspector an affidavit, duly executed by a company official, stating the specifications to which each chemical, used in the developer, conforms.

3.9 **Workmanship.** Workmanship shall be in accordance with high grade commercial practice covering the manufacture and packaging of photographic chemicals.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Materials and components.* The supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2 **Inspection lot.**

4.2.1 *Unit packages.* For the purpose of sampling the completed unit packages, a lot shall consist of all the unit packages offered for delivery at one time.

4.2.2 *Developer powder.* For the purpose of sampling the developer powder, a lot shall consist of all the chemicals packaged from the same batch and offered for delivery at one time. A batch shall consist of that quantity of chemicals prepared as one integral mix.

4.3 **Sampling.** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated herein.

4.4 **Examination of unit package.** The examination of the unit package shall be made in accordance with the classification of defects, inspection levels and acceptance levels (AQL's) set forth below.

4.4.1 *Appearance, workmanship, and marking.* Examine the unit package for defects in appearance, construction, workmanship, closure and markings. For determining sample size, the inspection level shall be I and the AQL shall be 1.5.

Examine	Defects
Appearance, construction and workmanship	Cracked, crushed, or any defects affecting serviceability.
Closure	Any leakage or seepage of contents. Closures not as specified.
Marking	Label data omitted, illegible, incorrect, incomplete, or not in accordance with 3.6.5.

4.4.2 *Closure.* Examine the closure of the unit package to determine that it is waterproof and airtight. For determining sample size, the inspection level shall be S-2 and the AQL shall be 4.0. Samples used for this examination may be used for subsequent examinations.

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Examine	Defects
Conduct the leakage test specified in PPP-C-96.	Leakage indicated by escaping air.

4.4.3 *Inner contents.* Examine the unit package contents for the appearance of the powder, inside of the outer container, construction, closure and workmanship of the inner container, and net weight. For determining sample size, the inspection level shall be S-2 and the AQL shall be 4.0

Examine	Defects
Space utilization of containers. Not required if waived, per 3.6.3.	Less than 80 percent of internal space utilized.
Net contents.	Not in accordance with 3.3.1.
Appearance of powder(s).	Not in powder form. Not homogeneous.
	Contains lumps, not free flowing
	Presence of foreign matter.
	Any discoloration or other evidence or deterioration.
	Any leakage or seepage of contents of inner container.
Appearance, construction and workmanship of inner container.	Cracked, crushed, or any defects affecting serviceability. Not in container as specified. Not clean.
Closure of inner container.	Any leakage or seepage of contents. Closures not as specified.
Marking of inner container.	Label or marking omitted, illegible, incorrect, incomplete.
Inside coating of outer container.[1]	Missing, does not completely cover inner surfaces, flaked or chipped.

[1] Contractor shall furnish a certificate of compliance that inner coating will not react physically or chemically with the contents of the container.

4.5 **Examination of developer powder.** Each batch lot of developer powder shall be subjected to the following tests. The sample shall consist of two unit packages selected at random from the lot. Failure to pass one or more tests shall be cause for rejection of the lot.

4.5.1 *Solubility test.* Verify the free flowing characteristics of the developer powder from one of the sample unit packages. Prepare a developer solution, using distilled water, from this powder according to the directions for mixing on the container. The developer powder shall dissolve readily without excessive stirring. Prepare a control solution, using distilled water, according to the formula specified in table I. Components of the control solution shall be weighed with an accuracy of plus or minus one percent. Cool both solutions to 68°F. and examine the test solution for clarity, dirt, and insoluble precipitates. Compare the color of the test solution with the color of the freshly mixed control solution. Cool the test solution to 60°F. There shall be no precipitation of any of the chemicals. The solutions prepared for this test may be used for subsequent tests.

4.5.2 *Alkalinity test.* Using the solutions prepared for the solubility test, measure the alkalinity of the test developer solution and the control developer solution with a properly calibrated pH meter, using an alkaline type glass electrode with the reference electrode. The alkalinity of the test solution shall be within plus or minus 0.1 pH unit of the control solution.

4.5.3 *Photographic test.* Expose identical sensitometric images, containing twenty-one steps, with a range in log exposure of 0.0 to 3.0, onto two sheets of medium speed, panchromatic, general purpose film of the same emulsion number. Using the solutions prepared for the solubility test, process one sheet of film in the test developer solution and one sheet in the control developer solution in accordance with the procedure described in ASA PH4.1, Manual Processing of Black-and-White Photographic Films and Plates, Methods for. Develop each of the films to a gamma of 0.9 plus or minus 0.1. The temperature of all solutions used in processing the films shall be 68° ±1/2°F. After drying, the densities of both sheets of film shall be measured with a densitometer. The densities, including base plus fog, of comparably exposed steps in each sheet shall agree within 0.04 from 0.1 density up

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to 0.60, 0.08 from 0.60 density up to 1.80 and 0.15 from 1.80 density up to 2.70. A lower fog level in the film processed in the test solution shall not be cause for rejection.

4.5.4 *Chemical test.* Using analytical methods with an accuracy of plus or minus 0.5 grams per liter for sodium sulfite and plus or minus 0.1 gram per liter for the balance of the chemical components, conduct a quantitative chemical analysis of the contents of the other sample unit package. Determine that the weight of each ingredient is within the tolerance specified in table I.

4.6 Storage life test. Two unit packages shall be selected from each batch lot and stored, undisturbed, at 120° ± 3°F. for a period of four weeks. The samples shall then be cooled to room temperature and one sample unit package shall then be examined. The developer powder shall not be dark in color, shall be free of dark specks, shall display no evidence of lumpiness or caking, shall not have been contaminated by the container(s) and shall readily dissolve. There shall be no corrosion of the container(s). The test developer made from the stored powder shall pass the photographic test of 4.5.3. In the photographic test, the time required for the test developer to develop the exposed film to a gamma of 0.9 plus or minus 0.1 shall not be more than one minute longer than the time required for the control solution to develop film to the same gamma. Failure of the sample unit package to pass the photographic test shall be cause for rejection of the entire batch. If the sample unit package fails to pass the other portions of the storage test, the second unit package shall be examined. Failure of the second unit package to pass this test shall be cause for rejection of the entire batch.

4.7 Test data. The contractor shall keep the test data specified herein, and a copy shall be made available to the Government inspector.

a. Test data shall be kept on each of the individual bulk chemicals used in the formula to insure that each chemical meets the chemical quality requirements. In lieu of such data, the contractor shall keep on file a certification and an average analysis from the manufacturer of each bulk chemical to prove that the chemicals supplied were in accordance with the specification.

b. Test data shall be kept on each batch of developer powder to assure that it has been accurately compounded.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the levels of preservation, packaging and packing protection used are defined in Fed. Std. No. 102).

5.1 Unit package. The developer powder shall be packaged in accordance with the requirements of section 3.

5.1.1 *Intermediate package.* The developer powder unit packages shall be overpackaged in intermediate containers conforming to PPP-B-636, type CF, class domestic, of any style. The intermediate container shall hold no more than 24 one gallon or 12 three and one half gallon sizes of unit packages, as applicable.

5.2 Packing.

5.2.1 *Level A.* Intermediate packages of developer powder shall be packed for shipment according to the requirements of PPP-P-16, (see 6.2).

5.2.2 *Level C.* Intermediate packages of developer powder shall be packed for shipment in exterior type shipping containers in a manner that will insure safe transportation to the point of delivery. Containers shall meet interstate Commerce Commission Regulations and Regulations of common carriers as applicable to the mode of transportation.

5.3 Marking and labeling.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the packages and shipping containers shall be

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marked in accordance with Fed. Std. No. 123. "The marking shall include the manufacturer's batch number and date of manufacture".

5.3.2 *Military activities.* In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129. "The marking shall include the manufacturer's batch number and date of manufacture".

6. NOTES

6.1 Intended use. The developer powder specified is intended for use in preparing the working solution of a general purpose developer for producing photographic negatives of moderate contrast.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size of the unit package and number of units required (see 1.2).
(*c*) Responsibility for inspection (see 4.1).
(*d*) Selection of applicable levels of packing (see 5.2.1 and 5.2.2).
(*e*) Marking and labeling (see 5.3 and 5.3.1).
(*f*) Data when required (see 3.2, 3.6.3, 3.8, 4.4.3 and 4.7).

6.3 Variety of limitations. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Chemicals, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Chemicals, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

CUSTODIANS:

Army—EL

Navy—WP

Air Force—67

Preparing Activity:

Navy—Wep

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